Executive Analysis Report

E-commerce performance, January 2024 to December 2025 · prepared for the Head of Operations · data: 340,795 transaction records

NET REVENUE 2025	GROSS PROFIT	GROSS MARGIN	ORDERS	AVG ORDER VALUE
€12.46M +29.2% vs 2024	**€4.92M** +30.0% vs 2024	**39.5%** +0.3pp vs 2024	**79,966** +40.3% vs 2024	**€155.78** –7.9% vs 2024

Executive summary

The business grew strongly and profitably. Net revenue rose 29.2% to €12.46M and gross profit 30.0% to €4.92M. Gross margin held at 39.5% (+0.3pp) even though shipping costs rose from 4.9% to 5.7% of revenue, because list-price increases offset the more expensive courier.

Revenue is a misleading guide to which products matter. 72 high-revenue products earn 26% of revenue but only 18% of gross profit, at a 28% margin against 44% for the best products. Duffel Bag Plus F951 ranks #20 by revenue but #44 by profit.

Discount codes did not pay for themselves. Discounted orders contain the same number of items as full-price orders but earn €23 less gross profit each: about €656k given away over two years with no measurable gain in volume or loyalty.

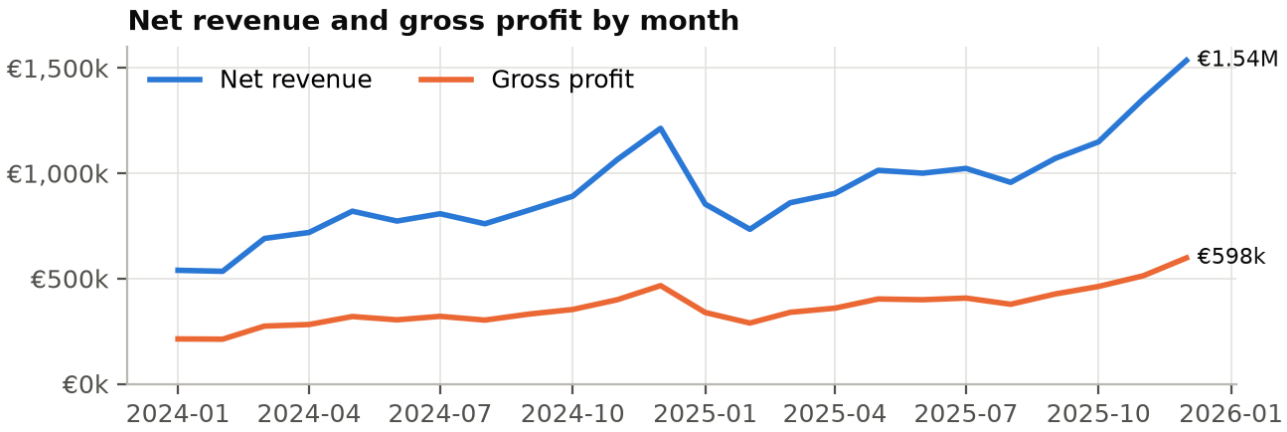
Customers come back far more often since the loyalty programme. The share of new customers who order again within 90 days rose from 27.8% to 41.3%: +11.3pp from the programme and +2.1pp from faster delivery.

Stock availability is the most urgent operational risk. 47 products need reordering today, 9 of them A-class best-sellers, with €64k of revenue at risk in January. In 2025, 12 A-class products ran out three or more times, costing an estimated €133k in lost sales.

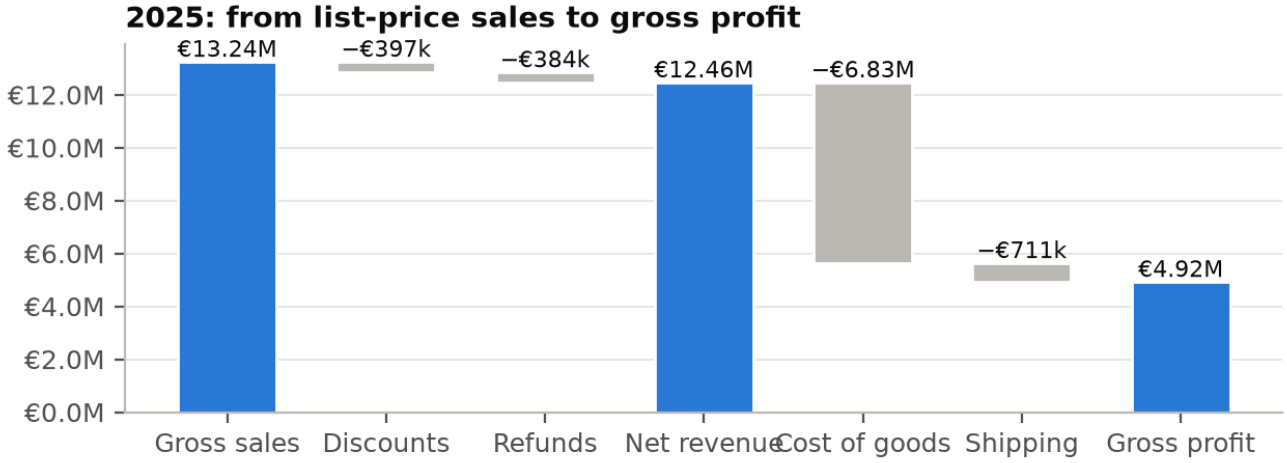
Top recommendations

- Order the 47 at-risk products today and raise safety stock on the 12 repeatedly out-of-stock A-class products.
- Rank products by gross profit, not revenue, in category reviews; re-price or re-source the volume-trap products.
- Stop blanket discount codes; test targeted offers against a hold-out group before rolling them out.
- Keep the loyalty programme and target the 10.4% of customers in the At Risk / Cannot Lose Them segments (16.8% of revenue) with a win-back campaign.

1. Business performance



Revenue grew in every quarter, and growth was not a Black Friday effect: rolling 90-day revenue was up between 26% and 28% year on year in both halves of 2025. Orders grew faster (40.3%) than revenue, so the average order value fell from €169.16 to €155.78. That fall is **product mix, not pricing**: the same products cost 2.1% more in 2025, but cheaper categories grew from 10.9% to 14.5% of revenue.

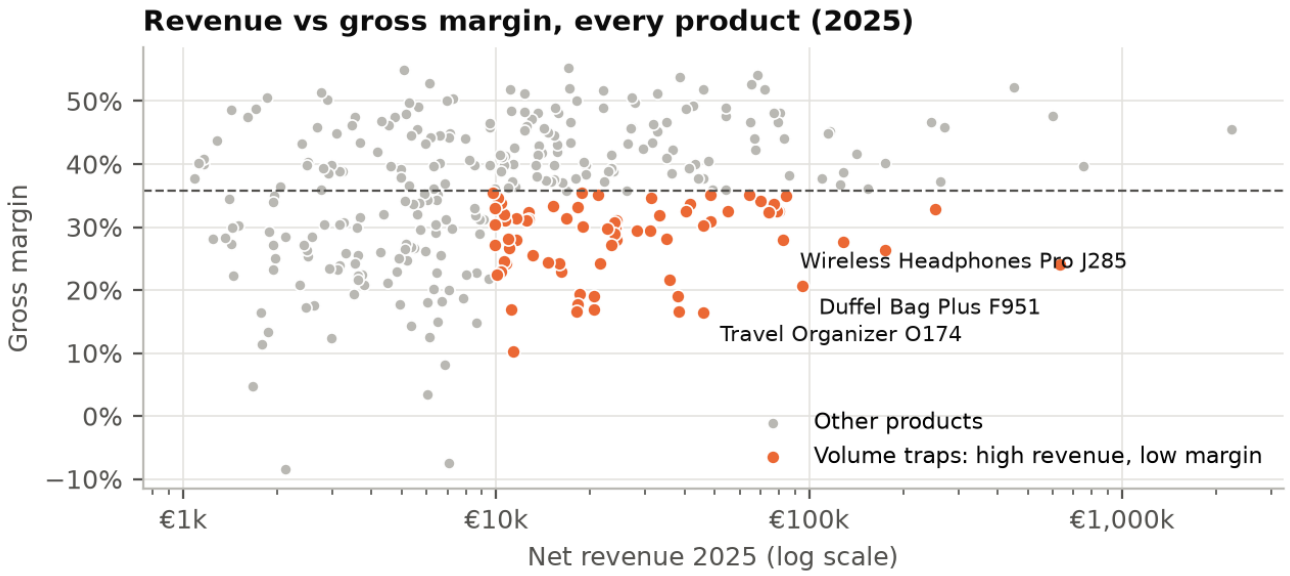


Of every €100 of list-price sales in 2025, €3 went in discounts and €3 in refunds. Cost of goods takes 55% of net revenue and shipping 5.7%, leaving a gross margin of 39.5%. Shipping is the cost growing fastest: the faster courier for Greece costs €1.38 more per order (€6.27 to €7.65), in exchange for on-time delivery rising from 37% to 68%.

Recommendation: report gross profit next to revenue in the monthly pack, and AOV next to units per order, so that a falling AOV is not mistaken for discounting.

2. Products and profitability

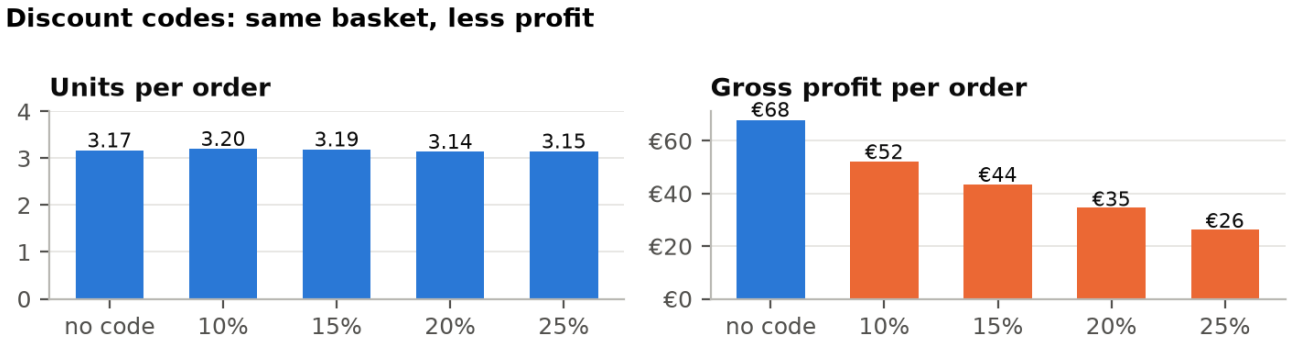
The product review used to rank products by revenue. After adding unit cost, discounts, refunds and the shipping each product causes, that ranking turned out to be misleading, so gross profit is now the KPI for product decisions.



72 products sit in the 'volume trap' corner: above-median revenue, below-median margin. They are 26% of 2025 revenue but 18% of gross profit. Two causes dominate: a high unit cost (low markup) and, for bulky items, shipping. Travel Organizer O174 drops from #50 by revenue to #96 by profit; Home & Kitchen and Fitness lose 19% and 14% of their revenue to shipping alone. 5 products lost money on every sale in 2025.

Product	Revenue rank	Profit rank	Margin	Main reason
Travel Organizer O174	#50	#96	16%	Low markup (high unit cost)
Duffel Bag Plus F951	#20	#44	21%	Low markup (high unit cost)
Wireless Headphones Pro J285	#24	#40	28%	Low markup (high unit cost)
Monitor Arm J285	#48	#64	30%	Low markup (high unit cost)
Game Controller H137	#45	#58	31%	Low markup (high unit cost)
Tea Kettle Mini M988	#14	#25	28%	Bulky: shipping eats the margin

Did discounting pay for itself?

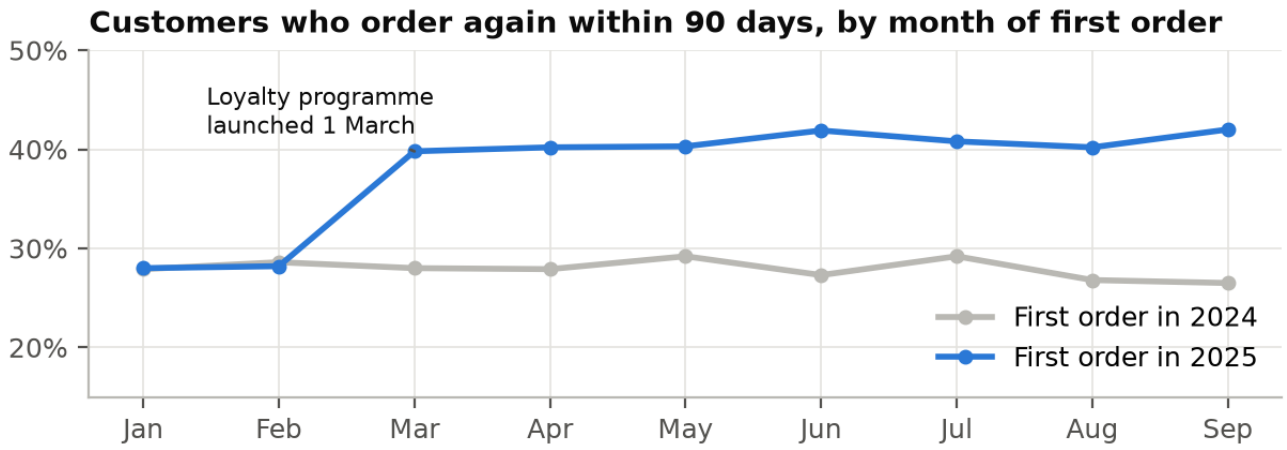


About one order in five used a 10-25% code. Compared within the same month (codes cluster in November-December), discounted orders held no more items (difference +0.008 (95% CI -0.017 to +0.034) items per order), and first-time buyers who used a code were no more likely to return. At a 41% margin a 15% code needs baskets ~57% larger to break even; the observed difference is under 1%.

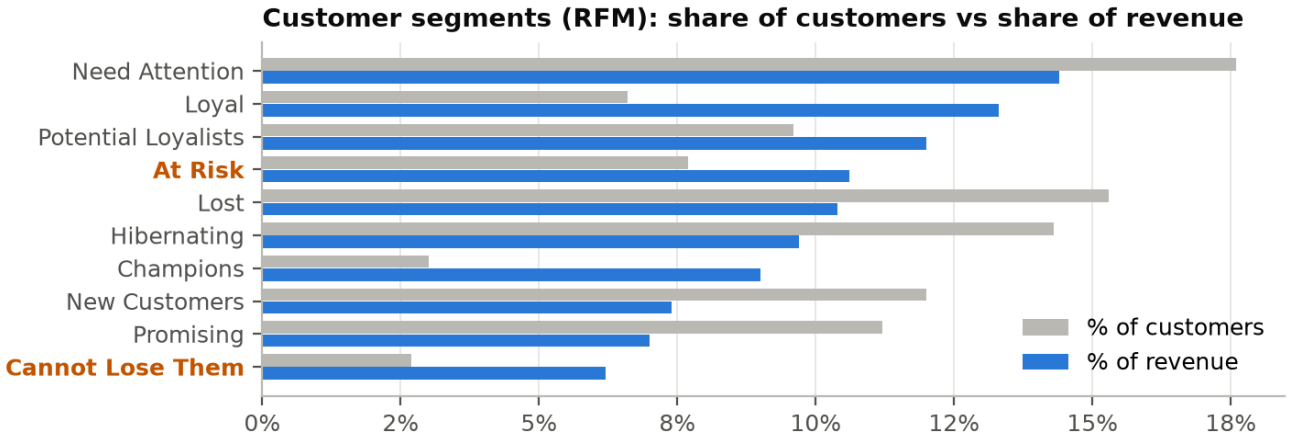
Recommendation: review price or supplier cost for the volume-trap products, add a minimum order value or shipping charge for bulky low-price items, and replace blanket codes with targeted offers tested against a hold-out group.

3. Customer insights

34.3% of customers have ordered more than once, and they generate 57.2% of revenue. Most who return do so quickly (median 36 days between orders), so the first weeks after a first order matter most.



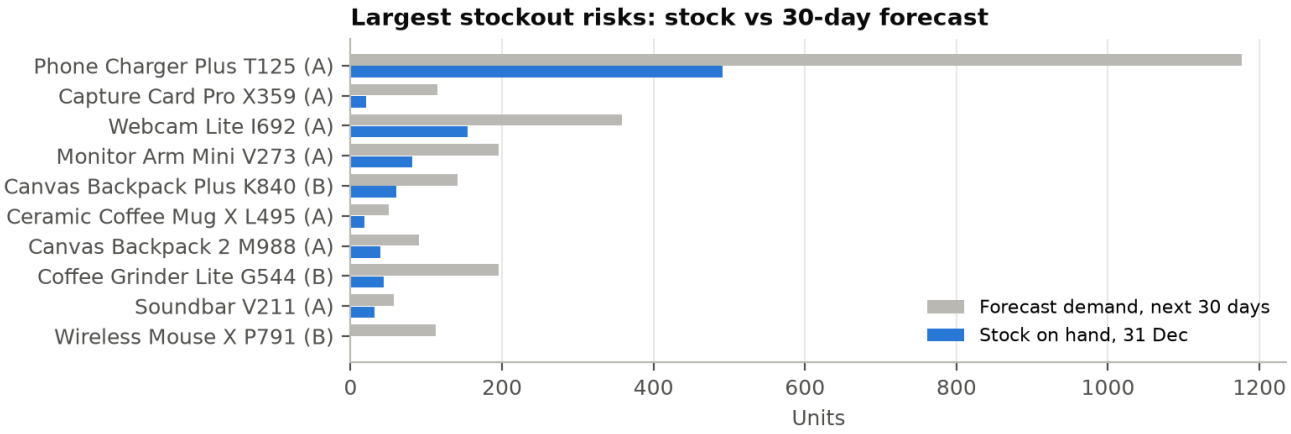
Repeat purchasing stepped up in March 2025, the month the loyalty programme launched, and stayed up. Comparing the same calendar months a year apart rules out seasonality. A faster courier for Greece arrived in May; separating the two, the programme explains 11.3pp of the 13.5pp improvement and faster delivery 2.1pp. Late deliveries matter: repeat 38.2% on time vs 25.9% late; on-time buyers have 1.77x the odds of returning.



At Risk and Cannot Lose Them customers have bought repeatedly but have gone quiet. They are 10.4% of customers but 16.8% of historical revenue, a better target than the larger Lost group, most of whom bought once.

Recommendation: keep the loyalty programme, add a second-order offer within 30 days of the first, and run a win-back campaign for At Risk / Cannot Lose Them before spending on lapsed one-time buyers.

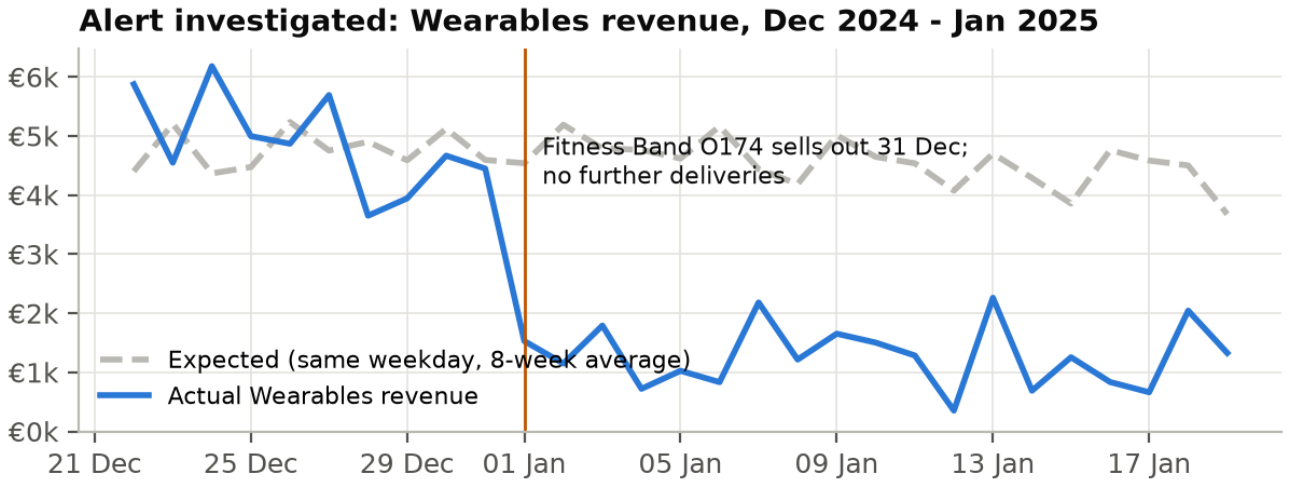
4. Product and inventory findings



SKU	Product	Class	Stock	30-day forecast	Difference	Action
SKU-00119	Phone Charger Plus T125	A	491	1,177	-686	Order today
SKU-00177	Capture Card Pro X359	A	20	115	-95	Expedite
SKU-00062	Webcam Lite I692	A	155	358	-203	Order today
SKU-00075	Monitor Arm Mini V273	A	82	195	-113	Order today
SKU-00286	Canvas Backpack Plus K840	B	60	141	-81	Order today
SKU-00233	Ceramic Coffee Mug X L495	A	18	50	-32	Expedite
SKU-00290	Canvas Backpack 2 M988	A	39	90	-51	Order today
SKU-00210	Coffee Grinder Lite G544	B	44	195	-151	Expedite

Demand for the next 30 days was forecast for every product from its recent sales on in-stock days, adjusted for the seasonal January drop. Tested on the previous four months, the forecast was more accurate than 'same as last month' every time (average error 15% vs 19%). 47 products will run out before or soon after a new delivery could arrive. Separately, 32 products hold €158k of stock (19% of stock value) with more than 120 days of cover.

An alert, investigated



Automated monitoring flagged Wearables revenue on 6 January 2025 at 84% below expectation. The investigation traced 98% of the drop to one product, Fitness Band O174, the category's best-seller (€956k in 2024). It sold out on 31 December 2024, its last delivery had been on 07 December, and it received no stock at all in 2025. Demand did not disappear; the product stopped being bought in.

Recommendation: place the reorders above today; raise safety stock on the A-class products that repeatedly run out; pause purchasing for overstocked lines; and confirm with purchasing whether Fitness Band O174 was discontinued on purpose. Add a daily alert when an A-class product's stock falls below its lead-time demand.

5. Recommendations

Ranked by expected impact and how quickly they can be acted on:

#	Action	Evidence	Owner
1	Reorder 47 at-risk products now; raise safety stock on 12 A-class SKUs	€64k at risk in January; €133k lost in 2025	Purchasing
2	Use gross profit, not revenue, to rank products; fix the volume traps	72 products: 26% of revenue, 18% of profit	Category mgmt
3	Replace blanket discount codes with targeted, tested offers	Same basket size, ~€656k profit given away	Marketing
4	Keep the loyalty programme; add a 30-day second-order offer	+11.3pp 90-day repeat rate	Marketing
5	Win back At Risk / Cannot Lose Them customers	10.4% of customers, 16.8% of revenue	CRM
6	Bring delivery abroad up to the Greek standard	Late first delivery lowers repeat rate	Operations
7	Release stock from 32 overstocked SKUs	€158k tied up	Purchasing

6. Methodology and limitations

Data. 340,795 transaction records exported from the order system, plus customers, products and daily stock counts. 48% of records needed a correction (duplicates, four date formats, inconsistent country names, missing values, impossible prices); each defect has a documented decision, and only 2,613 records were removed. After cleaning, the warehouse passes all 15 automated quality rules (completeness 98.1%, validity 100.0%, duplicates 0.0%).

Definitions. Net revenue = sales after discounts and refunds, excluding cancelled orders. Gross profit = net revenue minus cost of goods and outbound shipping (delivery is free for customers). Each KPI is defined once in the database and reused by every report.

Methods. SQL for all KPIs; significance tests for every claim that compares groups; same-weekday z-scores for anomaly detection; a seasonal, stockout-aware demand forecast tested against a naive baseline.

Limitations. The data is simulated, with realistic defects and business events built in: the methods carry over, the figures do not. Before/after comparisons (loyalty programme, courier) are observational; a hold-out group would measure them directly. Website sessions are not in the data, so the effect of discounts on conversion cannot be measured. Lost-sales estimates assume demand while out of stock equals demand when in stock, so they are an upper bound. Purchase orders in transit are not in the export.

Full analysis, SQL and code: github.com/kgiannopoulou/fullstack-ecommerce